

Unit 10, Lesson 7: Solving Equations Involving the Distributive Property

Benchmark

MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical context or real-world context, where all terms are rational numbers.

Learning Target

- ☐ I can solve two-step equations involving the distributive property.

10.7.1 Warm-Up: Two-Step Equations

1. Create an equation with the largest possible value for x by filling in the blue boxes using the digits 1 through 9. Each digit can only be used one time, at most. This may take several attempts. Show your work for each attempt you make.

$$\boxed{}x - \boxed{} = \boxed{}$$

10.7.2 Refresher: The Distributive Property

1. Several expressions are shown.

$$(-2 \cdot x) + 3$$

$$(x + 3)(-2)$$

$$-2x - 6$$

$$(-2) \cdot x + (-2) \cdot 3$$

$$(-2 + x)3$$

- a. Circle the expressions that are equivalent to $-2(x + 3)$.
- b. Use substitution to show that each of the expressions chosen is equivalent to $-2(x + 3)$.

2. Rewrite each expression by applying the distributive property.

a. $(-x + 6)4$

b. $-3.5(-7 + 2y)$

c. $\frac{7}{8}\left(16m - \frac{4}{5}\right)$

10.7.3 Collaboration: Equivalent Equations

1. All of the equations in the table below are equivalent. Work with your partner to explain how each equation is equivalent to the previous equation.

Equation	How is this equation equivalent to the equation above it?
$x = -6$	This is the given equation.
$x - 3 = -9$	
$-9 = x - 3$	
$900 = -100(x - 3)$	
$900 = -100x + 300$	

2. Two students wrote equations equivalent to $2\left(m + 1\frac{1}{4}\right) = 3.5$.

Simon’s Equation	Daphne’s Equation
$2m + 2\frac{1}{2} = 3.5$	$m + 1\frac{1}{4} = 1.75$

- a. Describe what Simon did to generate his equivalent equation.
- b. Describe what Daphne did to generate her equivalent equation.

10.7.4 Guided Instruction: Solving Equations Involving the Distributive Property

Equations can be solved in more than one way. When an equation is written in the form $a(b + c) = d$, it can be solved by either distributing a first or by dividing both sides the equation by a first.

1. The equation $65 = 8(9 - p)$ can be solved using either method.
- a. Solve the equation $65 = 8(9 - p)$ using each method.

Method 1 Apply the distributive property first	Method 2 Divide both sides by the factor being distributed first
$65 = 8(9 - p)$	$65 = 8(9 - p)$

- b. Verify the solution is correct using substitution.
- c. Discuss each method of solving the equation with your partner. Then, explain which method you preferred for solving this equation.

2. With your partner, decide who is Partner A and who is Partner B.
- a. Solve the equation $8\frac{1}{2} = \frac{2}{3}(x + 6)$ using the method indicated.

Partner A Apply the distributive property first	Partner B Divide both sides by the factor being distributed first
$8\frac{1}{2} = \frac{2}{3}(x + 6)$	$8\frac{1}{2} = \frac{2}{3}(x + 6)$

- b. Share your solution and your work with your partner. Correct any errors, if necessary. Then, write your partner’s method in the space provided once both partners agree the work is correct.
- c. Explain which method you preferred for solving this equation.

3. The perimeter of an equilateral triangle is 41.46 inches. If the length of each side is $(q - 5.7)$, the equation $3(q - 5.7) = 41.46$ can be used to find the value of q .
- a. What is the value of q ?
- b. The length of each side of the triangle is 13.82 inches because $41.46 \div 3 = 13.82$. Explain how the solution to the equation $3(q - 5.7) = 41.46$ can be used to show this as well.

10.7.5 Guided Practice: Solving Equations Involving the Distributive Property

Solve each equation using your preferred method. Then, verify that your solution is correct using substitution.

1. $\frac{3}{4}\left(k - 5\frac{1}{2}\right) = \frac{5}{8}$

2. $-3 = (w - 1.7)(-10)$

10.7.6 Collaboration: Error Analysis

Work with your partner to complete the following.

1. Francesca was asked to solve the equation $-4\frac{1}{2} = \frac{1}{3}(x - 1.5)$. Her work is shown.

$$-4\frac{1}{2} = \frac{1}{3}(x - 1.5)$$

$$-4.5 + 1.5 = \frac{1}{3}(x - 1.5) + 1.5$$

$$-2 = \frac{1}{3}x$$

$$\frac{-2}{\frac{1}{3}} = \frac{\frac{1}{3}x}{\frac{1}{3}}$$

$$x = -6$$

- Draw an arrow next to the step(s) where Francesca made a mistake.
 - Describe the mistake(s) Francesca made.
- c. Find the solution to $-4\frac{1}{2} = \frac{1}{3}(x - 1.5)$.

10.7.7 Wrap-Up: Ticket Out the Door

1. Solve $5\left(x + \frac{2}{5}\right) = -13$.